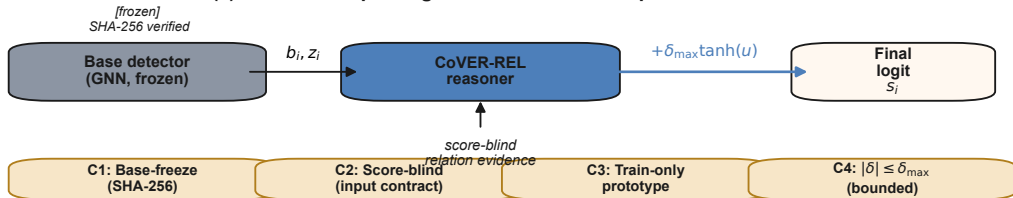
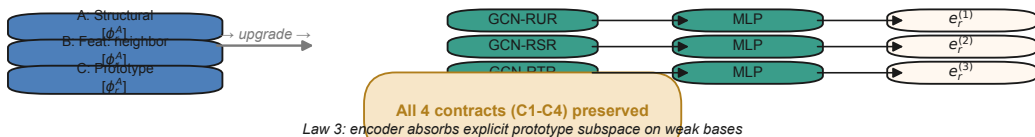


(a) C1 — RAER paradigm: contract-enforced post-hoc reasoner

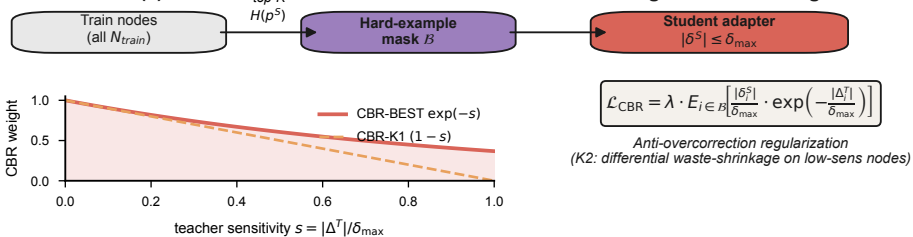


C1 hand-crafted evidence
(9-dim per relation)

(b) C2 — LREE: learnable per-relation evidence encoder



(c) C3 — Flash RAER + CBR-BEST loss: contract-budgeted residual regularizer



(d) Progressive AUPRC across the three contributions on 8 base $\times$ dataset cells

